Deep Learning Decision Support System for Automated Diabetic Retinopathy Screening and Grading.

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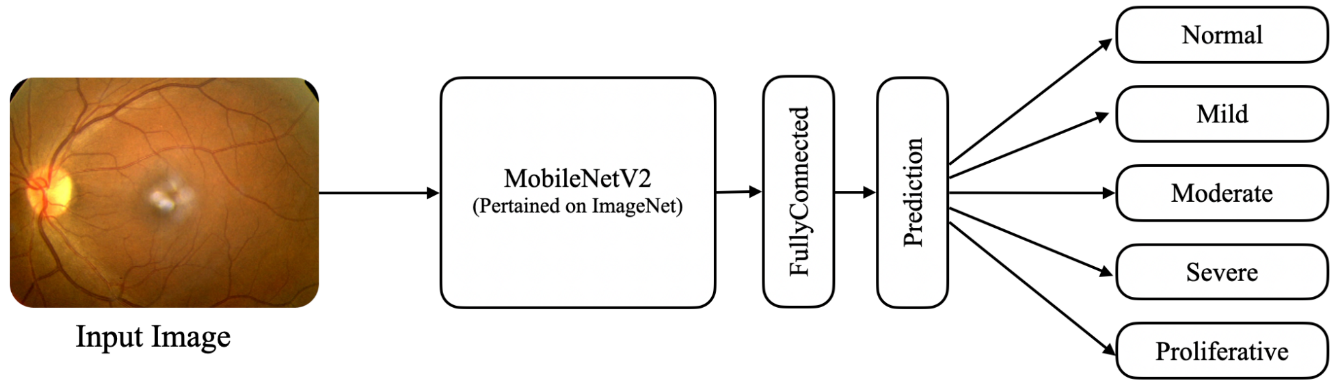
Problem & Motivation

- **Problem Context:** Diabetic retinopathy is a leading cause of irreversible blindness among working-age adults globally, and is a significant, yet largely unmanaged, complication of the growing diabetes epidemic in South Sudan.
- **Research Gap:** Reliance on multi-class classification models that excel at late-stage identification but underperform in detecting subtle, high-risk microvascular anomalies.
- **Objective:** To develop and test a classification model for detecting Diabetic Retinopathy from retinal images.

Methodology

System Architecture

Our system utilizes a lightweight CNN network with a ***MobileNetV2*** *backbone to classify retina images to five categories . We chose this architecture to enable seamless deployment to* ***edge devices****.*



Dataset

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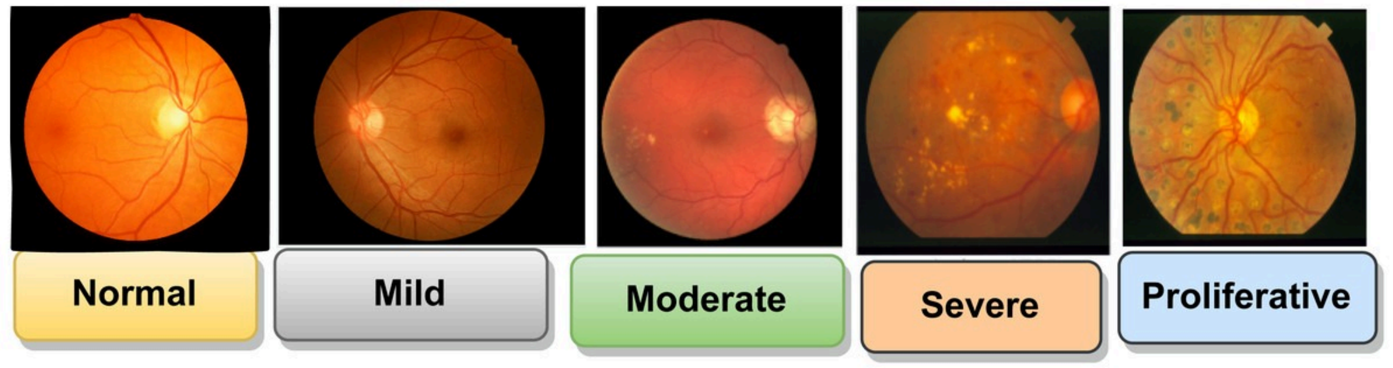
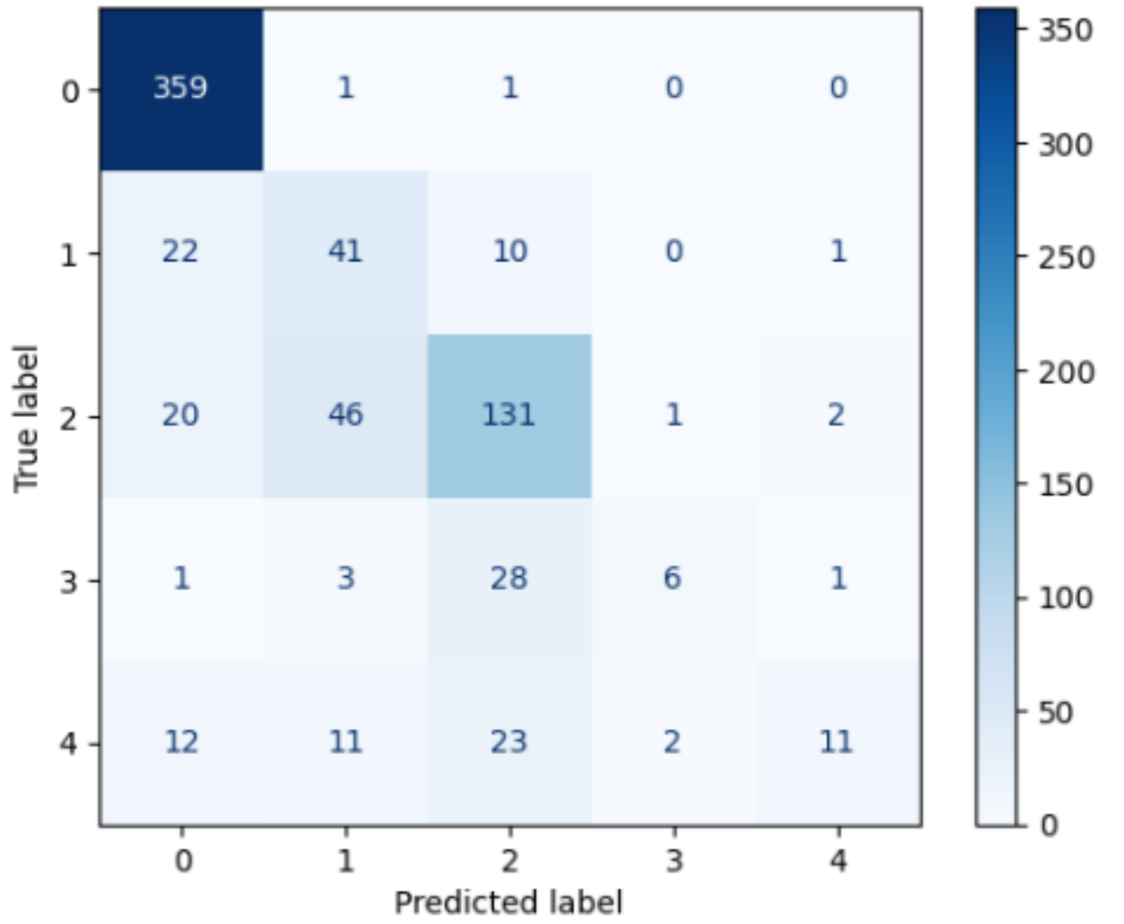
We use the AGAR300 dataset of fundus images sourced kaggle. We divided the data into 80% training and 20% testing.



Evaluation & Key Results

Performance

The system achieved an overall accuracy of 74% and a weighted F1-score of 0.70



Significance & Conclusion

- **Contribution:** A lightweight edge-enabled model for retinopathy screening..
- **Impact:** The system can assist ophthalmologists or optometrists in screening patients .
- **Future Work:** Clinical Integration: Moving from a backend "proof-of-concept" to a live hospital database integration. Also Fine-tuning the model to reach the original 85% target using advanced ensemble methods.